

COMP 2121 – Quiz 2

Time: 20 min.

Possible marks: 5

Name: SOLUTION

Student #:

1. Use rules of inference to prove that the following argument is valid.

Note: You must use the provided rules of inference and list the rule. No marks will be given for proofs that involve a truth table. (2.5 marks)

	STEPS	REASON
$y \wedge x$		
$y \rightarrow \neg(x \wedge w)$	1) $y \wedge x$	Given
$\neg z \rightarrow w$	2) y	CS 1
	3) $y \rightarrow \neg(x \wedge w)$	Given
$\therefore z$	4) $\neg(x \wedge w)$	MP 2,3
	5) $\neg x \vee \neg w$	DeMorgan's 4
	6) x	CS 1
	7) $\neg w$	DS 5,6
	8) $\neg z \rightarrow w$	Given
	9) z	MT 7,8
	10)	
	11)	
	12)	
	13)	

2. Prove if each of the following statements are TRUE or FALSE. Note that you must provide justification for your answers using a table of values. No marks will be given for answers that use other methods of justification or that provide no justification.

The universe for x is $\{-3, 0, 9\}$.

The universe for y is $\{-2, -1, 0, 1\}$

(a) (0.5 marks) $\forall x, \forall y, x > |3y|$

TRUE / FALSE (circle)

x	y	$x > 3y $
-3	0	-3 > 0 F

∴ False

(b) (1 mark) $\exists x, \forall y, x > |3y|$

TRUE / FALSE (circle)

x	y	$x > 3y $
9	-2	9 > 6 T
9	-1	9 > 3 T
9	0	9 > 0 T
9	1	9 > 3 T

∴ True

(c) (1 mark) $\forall x, \exists y, x > |3y|$

TRUE / FALSE (circle)

x	y	$x > 3y $
-3	-2	-3 > 6 F
-3	-1	-3 > 3 F
-3	0	-3 > 0 F
-3	1	-3 > 3 F

∴ False

